

Assignment

Date _____ Period _____

Simplify. Write your answer in rectangular form.

1) $4\left(\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3}\right) \cdot 6\left(\cos \frac{4\pi}{3} + i\sin \frac{4\pi}{3}\right)$

$r = 4 \cdot 6 = 24$

$\theta = \frac{2\pi}{3} + \frac{4\pi}{3} = \frac{6\pi}{3} = 2\pi$ full turn $\rightarrow 0$

$= 24(\cos 0 + i\sin 0)$

$= 24(1 + 0i)$

$\boxed{24}$

2) $6\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right) \cdot 4\left(\cos \frac{5\pi}{4} + i\sin \frac{5\pi}{4}\right)$

$r = 6 \cdot 4 = 24 \quad \theta = \frac{2\pi}{12} + \frac{15\pi}{12} = \frac{17\pi}{12}$

$= 24\left(\cos \frac{17\pi}{12} + i\sin \frac{17\pi}{12}\right)$

$\frac{17\pi}{12} = 255^\circ$, not on unit circle: $210^\circ + 45^\circ$, sum formula

$\cos 255^\circ = \cos 210^\circ \cos 45^\circ - \sin 210^\circ \sin 45^\circ$

$= \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(-\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2} - \sqrt{6}}{4}$

$\sin 255^\circ = \sin 210^\circ \cos 45^\circ + \cos 210^\circ \sin 45^\circ$

$= \left(-\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = \frac{-\sqrt{2} - \sqrt{6}}{4}$

$= 24\left(\frac{\sqrt{2} - \sqrt{6}}{4}\right) + 24i\left(\frac{-\sqrt{2} - \sqrt{6}}{4}\right)$

$\boxed{6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2})i}$

3) $(2 - i)(-2 + 5i)$

not unit-circle angles: $\tan^{-1}$ on calc (like HW #15), then product theorem

$2 - i: \quad r = \sqrt{2^2 + (-1)^2} = \sqrt{5} \quad \theta = \tan^{-1}\left(\frac{-1}{2}\right) = -26.5651^\circ, \text{ Q4} \rightarrow +360^\circ = 333.4349^\circ$

$-2 + 5i: \quad r = \sqrt{(-2)^2 + 5^2} = \sqrt{29} \quad \theta = \tan^{-1}\left(\frac{5}{-2}\right) = -68.1986^\circ, \text{ Q2} \rightarrow +180^\circ = 111.8014^\circ$

$(\sqrt{5} \text{cis } 333.4349^\circ)(\sqrt{29} \text{cis } 111.8014^\circ)$

$r = \sqrt{5} \cdot \sqrt{29} = \sqrt{145} \quad \theta = 333.4349^\circ + 111.8014^\circ = 445.2363^\circ - 360^\circ = 85.2363^\circ$

$= \sqrt{145}(\cos 85.2363^\circ + i\sin 85.2363^\circ)$

$= \sqrt{145}\cos 85.2363^\circ + \left(\sqrt{145}\sin 85.2363^\circ\right)i$

$\boxed{1 + 12i}$

$$4) \frac{-5 + 4i}{4 + 5i}$$

top: $-5 + 4i$

$$r = \sqrt{(-5)^2 + 4^2} = \sqrt{41}$$

$$\theta = \tan^{-1}\left(\frac{4}{-5}\right) = -38.6598^\circ, \text{ Q2} \rightarrow 141.3402^\circ$$

$$= \sqrt{41} \text{ cis } 141.3402^\circ$$

bottom: $4 + 5i$

$$r = \sqrt{4^2 + 5^2} = \sqrt{41}$$

$$\theta = \tan^{-1}\left(\frac{5}{4}\right) = 51.3402^\circ \text{ (Q1)}$$

$$= \sqrt{41} \text{ cis } 51.3402^\circ$$

$$\frac{\sqrt{41} \text{ cis } 141.3402^\circ}{\sqrt{41} \text{ cis } 51.3402^\circ} = 1 \text{ cis } 90^\circ$$

$$= \cos 90^\circ + i \sin 90^\circ = 0 + i(1)$$

$$\boxed{i}$$

$$5) \frac{4 + 3i}{2i}$$

top: $4 + 3i$

$$r = \sqrt{4^2 + 3^2} = 5$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.8699^\circ \text{ (Q1)}$$

$$= 5 \text{ cis } 36.8699^\circ$$

bottom: $2i = 0 + 2i$

$$r = 2, \theta = 90^\circ \text{ on the imaginary axis}$$

$$= 2 \text{ cis } 90^\circ$$

$$\frac{5 \text{ cis } 36.8699^\circ}{2 \text{ cis } 90^\circ} = \frac{5}{2} \text{ cis } (-53.1301^\circ)$$

$$+ 360^\circ: \frac{5}{2} \text{ cis } 306.8699^\circ$$

$$= \frac{5}{2} (\cos 306.8699^\circ + i \sin 306.8699^\circ)$$

$$= \frac{5}{2} (0.6) + \frac{5}{2} (-0.8)i$$

$$\boxed{\frac{3}{2} - 2i}$$

$$6) \frac{4 - 6i}{-6 - 2i}$$

top: $4 - 6i$

$$r = \sqrt{4^2 + (-6)^2} = \sqrt{52} = 2\sqrt{13}$$

$$\theta = \tan^{-1}\left(\frac{-6}{4}\right) = -56.3099^\circ, \text{ Q4} \rightarrow 303.6901^\circ$$

bottom: $-6 - 2i$

$$r = \sqrt{(-6)^2 + (-2)^2} = \sqrt{40} = 2\sqrt{10}$$

$$\theta = \tan^{-1}\left(\frac{-2}{-6}\right) = 18.4349^\circ, \text{ Q3} \rightarrow 198.4349^\circ$$

$$\frac{2\sqrt{13} \text{ cis } 303.6901^\circ}{2\sqrt{10} \text{ cis } 198.4349^\circ} = \frac{\sqrt{13}}{\sqrt{10}} \text{ cis } 105.2552^\circ$$

$$= \sqrt{1.3} (\cos 105.2552^\circ + i \sin 105.2552^\circ)$$

$$= -0.3 + 1.1i$$

$$\boxed{-\frac{3}{10} + \frac{11}{10}i}$$

$$7) (6(\cos 30 + i \sin 30))^4$$

$$6^4 (\cos 4(30^\circ) + i \sin 4(30^\circ))$$

$$= 1296 (\cos 120^\circ + i \sin 120^\circ)$$

$$= 1296 \left(-\frac{1}{2}\right) + 1296i \left(\frac{\sqrt{3}}{2}\right)$$

$$\boxed{-648 + 648\sqrt{3}i}$$

$$8) (5 - 3i)^4$$

$$r = \sqrt{5^2 + (-3)^2} = \sqrt{34}$$

$$\theta = \tan^{-1}\left(\frac{-3}{5}\right) = -30.9638^\circ, \text{ Q4} \rightarrow 329.0362^\circ$$

$$\left(\sqrt{34} \operatorname{cis} 329.0362^\circ\right)^4$$

$$= (\sqrt{34})^4 \operatorname{cis} 4(329.0362^\circ)$$

$$= 1156 \operatorname{cis} 1316.1448^\circ$$

$$1316.1448^\circ - 3(360^\circ) = 236.1448^\circ$$

$$= 1156 (\cos 236.1448^\circ + i \sin 236.1448^\circ)$$

$$\boxed{-644 - 960i}$$

$$9) (5 - 6i)^4$$

$$r = \sqrt{5^2 + (-6)^2} = \sqrt{61}$$

$$\theta = \tan^{-1}\left(\frac{-6}{5}\right) = -50.1944^\circ, \text{ Q4} \rightarrow 309.8056^\circ$$

$$\left(\sqrt{61} \operatorname{cis} 309.8056^\circ\right)^4$$

$$= (\sqrt{61})^4 \operatorname{cis} 4(309.8056^\circ)$$

$$= 3721 \operatorname{cis} 1239.2224^\circ$$

$$1239.2224^\circ - 3(360^\circ) = 159.2224^\circ$$

$$= 3721 (\cos 159.2224^\circ + i \sin 159.2224^\circ)$$

$$\boxed{-3479 + 1320i}$$

Find all n th roots. Write your answers in polar form.

$$10) 1 + i\sqrt{3}, n = 5$$

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \text{ (Q1)}$$

$$\left(2 \operatorname{cis} \frac{\pi}{3}\right)^{1/5} = \sqrt[5]{2} \operatorname{cis} \frac{\pi/3}{5} = \sqrt[5]{2} \operatorname{cis} \frac{\pi}{15}$$

$$+\frac{2\pi}{5} = +\frac{6\pi}{15} \text{ each time } \mathbf{5 \text{ roots}}$$

$$\boxed{\sqrt[5]{2} \operatorname{cis} \frac{\pi}{15}, \sqrt[5]{2} \operatorname{cis} \frac{7\pi}{15}, \sqrt[5]{2} \operatorname{cis} \frac{13\pi}{15},}$$

$$\boxed{\sqrt[5]{2} \operatorname{cis} \frac{19\pi}{15}, \sqrt[5]{2} \operatorname{cis} \frac{5\pi}{3}}$$

$$11) -\frac{5}{2} + \frac{5\sqrt{3}}{2}i, n = 5$$

$$r = \sqrt{\left(-\frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\theta = \tan^{-1}\left(\frac{5\sqrt{3}/2}{-5/2}\right) = \tan^{-1}(-\sqrt{3}), \text{ Q2} \rightarrow 120^\circ$$

$$(5 \operatorname{cis} 120^\circ)^{1/5} = \sqrt[5]{5} \operatorname{cis} \frac{120^\circ}{5} = \sqrt[5]{5} \operatorname{cis} 24^\circ$$

$$+\frac{360^\circ}{5} = +72^\circ \text{ each time } \mathbf{5 \text{ roots}}$$

$$\boxed{\sqrt[5]{5} \operatorname{cis} 24^\circ, \sqrt[5]{5} \operatorname{cis} 96^\circ, \sqrt[5]{5} \operatorname{cis} 168^\circ,}$$

$$\boxed{\sqrt[5]{5} \operatorname{cis} 240^\circ, \sqrt[5]{5} \operatorname{cis} 312^\circ}$$

$$12) \frac{5}{2} - \frac{5\sqrt{3}}{2}i, n=2$$

$$r = \sqrt{\left(\frac{5}{2}\right)^2 + \left(-\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\theta = \tan^{-1}\left(\frac{-5\sqrt{3}/2}{5/2}\right) = \tan^{-1}(-\sqrt{3}), \text{ Q4} \rightarrow 300^\circ$$

$$(5 \operatorname{cis} 300^\circ)^{1/2} = \sqrt{5} \operatorname{cis} \frac{300^\circ}{2} = \sqrt{5} \operatorname{cis} 150^\circ \quad + \frac{360^\circ}{2} = +180^\circ \quad 2 \text{ roots}$$

$$\sqrt{5} \operatorname{cis} 150^\circ, \quad \sqrt{5} \operatorname{cis} 330^\circ$$

Find all roots. Answer in standard form.

$$13) x^3 - 64 = 0$$

$$x^3 = 64 \quad x = (64 \operatorname{cis} 0^\circ)^{1/3} \quad 64 = 64 + 0i, \theta = 0^\circ$$

$$x = \sqrt[3]{64} \operatorname{cis} \frac{0^\circ}{3} = 4 \operatorname{cis} 0^\circ \quad +120^\circ \text{ each time: } 4 \operatorname{cis} 0^\circ, 4 \operatorname{cis} 120^\circ, 4 \operatorname{cis} 240^\circ$$

$$4 \operatorname{cis} 0^\circ = 4(1) + 4i(0) = 4$$

$$4 \operatorname{cis} 120^\circ = 4\left(-\frac{1}{2}\right) + 4i\left(\frac{\sqrt{3}}{2}\right) = -2 + 2i\sqrt{3}$$

$$4 \operatorname{cis} 240^\circ = 4\left(-\frac{1}{2}\right) + 4i\left(-\frac{\sqrt{3}}{2}\right) = -2 - 2i\sqrt{3}$$

$$\{4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3}\}$$

Answers to Assignment (ID: 1)

1) 24

4) i

8) $-644 - 960i$

2) $6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2}) \cdot i$

5) $\frac{3}{2} - 2i$

9) $-3479 + 1320i$

6) $-\frac{3}{10} + \frac{11}{10}i$

10) $\sqrt[5]{2} \left(\cos \frac{\pi}{15} + i \sin \frac{\pi}{15} \right)$
 $\sqrt[5]{2} \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right)$
 $\sqrt[5]{2} \left(\cos \frac{13\pi}{15} + i \sin \frac{13\pi}{15} \right)$
 $\sqrt[5]{2} \left(\cos \frac{19\pi}{15} + i \sin \frac{19\pi}{15} \right)$
 $\sqrt[5]{2} \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$

3) $1 + 12i$

7) $-648 + 648i\sqrt{3}$

11) $\sqrt[5]{5}(\cos 24 + i \sin 24)$
 $\sqrt[5]{5}(\cos 96 + i \sin 96)$
 $\sqrt[5]{5}(\cos 168 + i \sin 168)$
 $\sqrt[5]{5}(\cos 240 + i \sin 240)$
 $\sqrt[5]{5}(\cos 312 + i \sin 312)$

12) $\sqrt{5}(\cos 150 + i \sin 150)$
 $\sqrt{5}(\cos 330 + i \sin 330)$

13) $\{4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3}\}$